

Predictive Modelling

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3: ARMA models

Outline

Autoregression (of order p)

Partial autocorrelation function

Moving average model (of order q)

ARMA(p, q)

ARIMA

Seasonal ARIMA

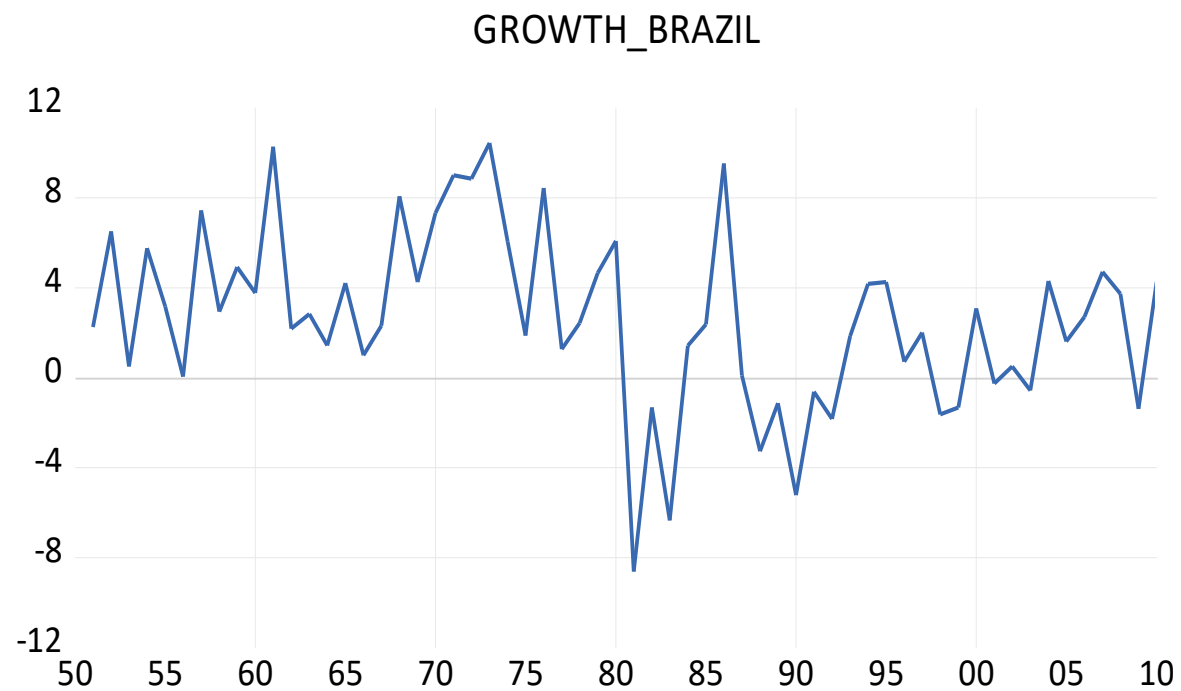
Autoregression of order p

An AR(1) model implies a specific pattern for the (theoretical) autocorrelations

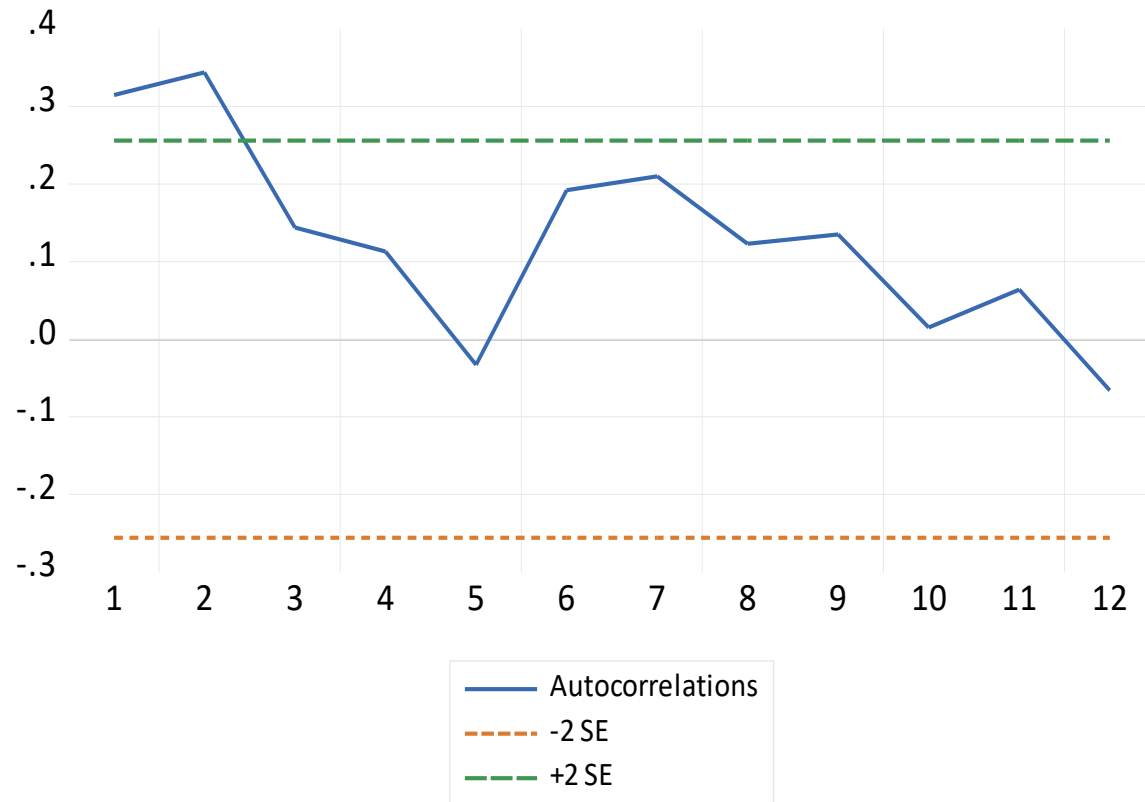
Recall

$$\begin{aligned}\rho_1 &= \phi_1 \\ \rho_2 &= \phi_1^2 \\ \rho_3 &= \phi_1^3 \\ &\dots \\ \rho_k &= \phi_1^k\end{aligned}$$

AR(1) may not suffice



Autocorrelations, GDP growth Brazil



Autoregression of order 2

$$y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} + \varepsilon_t$$

Again, theoretical autocorrelations can be derived. Here,

$$\rho_1 = \frac{\phi_1}{1 - \phi_2}$$

$$\rho_2 = \frac{\phi_1^2}{1 - \phi_2} + \phi_2$$

$$\rho_k = \phi_1 \rho_{k-1} + \phi_2 \rho_{k-2}$$

Not very useful in practice to easily identify this model.

Partial autocorrelation function

Run K auxiliary regressions

$$\begin{aligned}y_t &= \pi_1 y_{t-1} + u_t \\y_t &= \lambda_1 y_{t-1} + \pi_2 y_{t-2} + u_t \\y_t &= \lambda_1 y_{t-1} + \lambda_2 y_{t-2} + \pi_3 y_{t-3} + u_t \\y_t &= \lambda_1 y_{t-1} + \lambda_2 y_{t-2} + \lambda_3 y_{t-3} + \pi_4 y_{t-4} + u_t\end{aligned}$$

$$y_t = \lambda_1 y_{t-1} + \lambda_2 y_{t-2} + \lambda_3 y_{t-3} + \dots + \pi_K y_{t-K} + u_t$$

The estimated $\pi_1, \pi_2, \pi_3, \pi_4, \dots, \pi_K$ are the first K partial autocorrelations

Significance?

The standard error of partial autocorrelations is also

$$\frac{1}{\sqrt{T}}$$

Taking a 5% confidence level, then π_k is significant when

$$\hat{\pi}_k > \frac{2}{\sqrt{T}} \text{ or } \hat{\pi}_k < -\frac{2}{\sqrt{T}}$$

Identification of an $AR(p)$

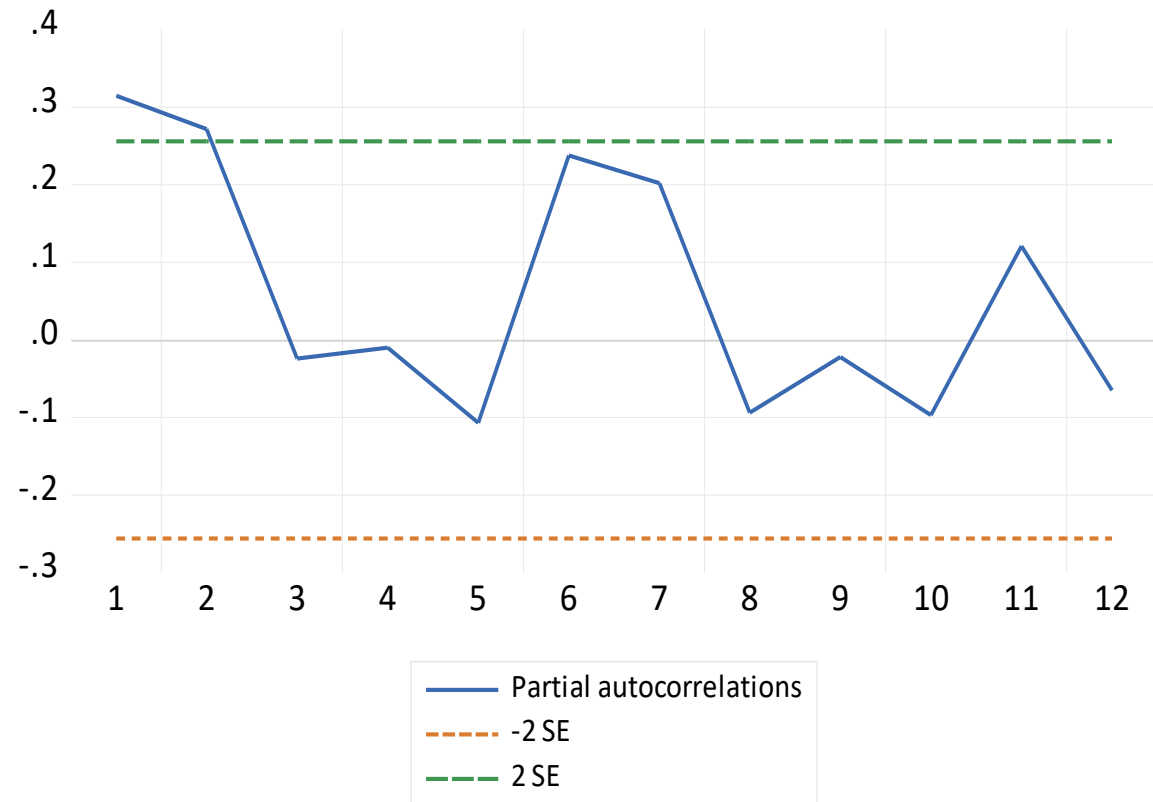
An autoregression of order p may be a good guess for $g(\cdot)$ when

$$\hat{\pi}_p > \frac{2}{\sqrt{T}} \text{ or } \hat{\pi}_p < -\frac{2}{\sqrt{T}}$$

While at the same time

$$-\frac{2}{\sqrt{T}} < \hat{\pi}_{p+1} < \frac{2}{\sqrt{T}}$$

Partial autocorrelations, growth Brazil



Estimation results AR(2) (OLS)

$$\textit{Growth Brazil}_t = 1.239 + 0.228 \textit{Growth Brazil}_{t-1} + 0.277 \textit{Growth Brazil}_{t-2}$$

with standard errors 0.625, 0.129 and 0.130, respectively

Autoregression of order p

$$y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} + \phi_3 y_{t-3} + \dots + \phi_p y_{t-p} + \varepsilon_t$$

Parameters can be estimated using Ordinary Least Squares (with t tests and p values the usual way)

Subset autoregression of order p (for example $p = 12$):

$$y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} + \phi_{12} y_{t-12} + \varepsilon_t$$

And this is also an AR(p):

$$y_t = \phi_p y_{t-p} + \varepsilon_t$$

Note the two variants

$$y_t = \delta + \phi_1 y_{t-1} + \varepsilon_t$$

versus

$$y_t = \mu + u_t$$

with

$$u_t = \phi_1 u_{t-1} + \varepsilon_t$$

Differences become relevant when model is for example

$$y_t = \delta + \phi_1 y_{t-1} + \beta x_t + \varepsilon_t$$

versus

$$y_t = \mu + \alpha x_t + u_t$$

with

$$u_t = \phi_1 u_{t-1} + \varepsilon_t$$

AR(1) with extra variable

$$y_t = \mu + \alpha x_t + u_t$$

with

$$u_t = \phi_1 u_{t-1} + \varepsilon_t$$

Becomes (because $u_t = y_t - \mu - \alpha x_t$)

$$(y_t - \mu - \alpha x_t) = \phi_1 (y_{t-1} - \mu - \alpha x_{t-1}) + \varepsilon_t$$

or

$$y_t = (1 - \phi_1)\mu + \phi_1 y_{t-1} + \alpha x_t - \alpha \phi_1 x_{t-1} + \varepsilon_t$$

Note the difference with

$$y_t = \delta + \phi_1 y_{t-1} + \beta x_t + \varepsilon_t$$

Moving average model of order 1

MA(1)

$$y_t = \varepsilon_t + \theta_1 \varepsilon_{t-1}$$

$$E(y_t) = E(\varepsilon_t) + \theta_1 E(\varepsilon_{t-1}) = 0$$

MA(1) with intercept

$$y_t = \mu + \varepsilon_t + \theta_1 \varepsilon_{t-1}$$

Now:

$$E(y_t) = \mu$$

Note the difference with AR(1)!

Autocorrelations MA(1)

$$\begin{aligned}\gamma_0 &= E(y_t^2) = E(\varepsilon_t + \theta_1 \varepsilon_{t-1})(\varepsilon_t + \theta_1 \varepsilon_{t-1}) = E(\varepsilon_t^2) + 2\theta_1 E(\varepsilon_t \varepsilon_{t-1}) + \theta_1^2 E(\varepsilon_{t-1}^2) \\ &= \sigma^2 + 0 + \theta_1^2 \sigma^2 = (1 + \theta_1^2) \sigma^2\end{aligned}$$

$$\begin{aligned}\gamma_1 &= E(y_t y_{t-1}) = E(\varepsilon_t + \theta_1 \varepsilon_{t-1})(\varepsilon_{t-1} + \theta_1 \varepsilon_{t-2}) \\ &= E(\varepsilon_t \varepsilon_{t-1}) + \theta_1 E(\varepsilon_t \varepsilon_{t-2}) + \theta_1 E(\varepsilon_{t-1}^2) + \theta_1^2 E(\varepsilon_{t-1} \varepsilon_{t-2}) = 0 + 0 + \theta_1 \sigma^2 + 0 = \theta_1 \sigma^2\end{aligned}$$

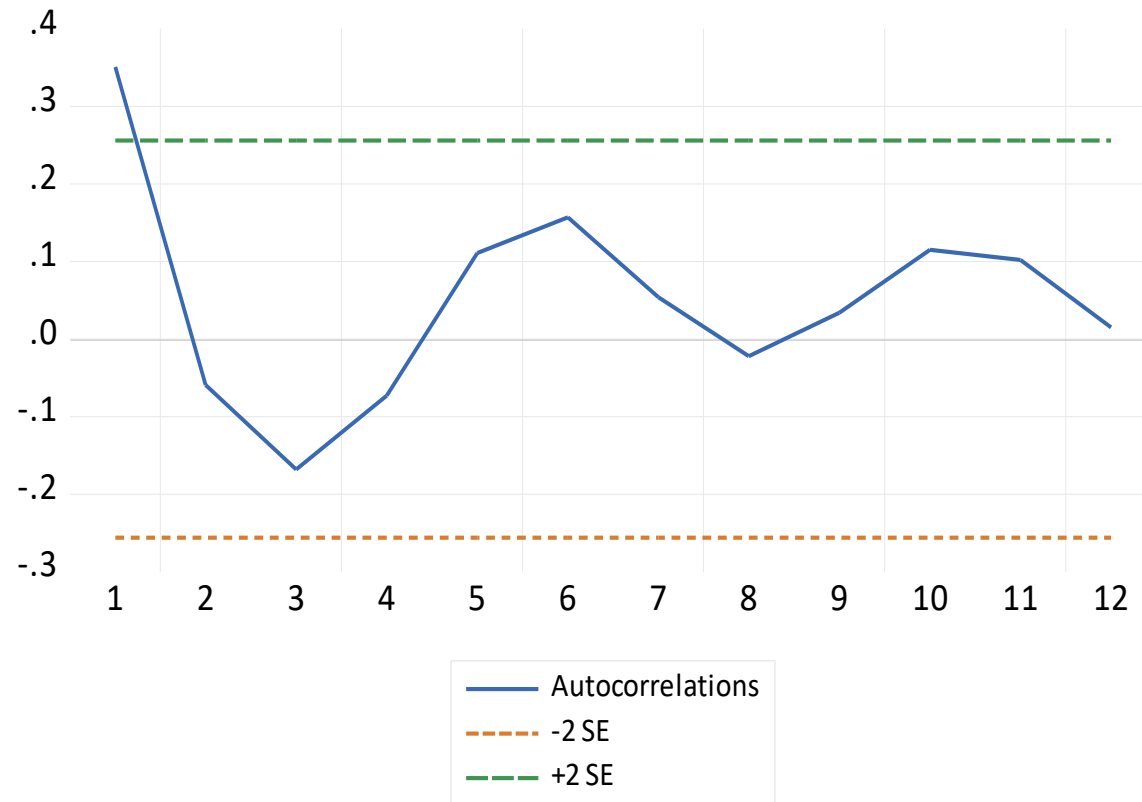
$$\gamma_2 = E(y_t y_{t-2}) = E(\varepsilon_t + \theta_1 \varepsilon_{t-1})(\varepsilon_{t-2} + \theta_1 \varepsilon_{t-3}) = 0$$

So,

$$\rho_1 = \frac{\theta_1 \sigma^2}{(1 + \theta_1^2) \sigma^2} = \frac{\theta_1}{(1 + \theta_1^2)}$$

$$\rho_k = 0 \text{ for } k = 2, 3, 4, \dots$$

Economic growth (GDP) in Peru



MA model of order q

$$y_t = \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \cdots + \theta_q \varepsilon_{t-q}$$

Parameters can be estimated using Iterative Least Squares or Maximum Likelihood (not OLS because the variables have to be estimated alongside)

Subset MA model of order q , for example

$$y_t = \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_q \varepsilon_{t-q}$$

or

$$y_t = \varepsilon_t + \theta_q \varepsilon_{t-q}$$

Autocorrelations of MA(q)

$$\rho_1 = f(\theta_1, \theta_2, \dots, \theta_q)$$

$$\rho_2 = f(\theta_1, \theta_2, \dots, \theta_q)$$

...

$$\rho_q = f(\theta_1, \theta_2, \dots, \theta_q)$$

And (!)

$$\rho_{q+1} = 0$$

$$\rho_{q+2} = 0$$

...

ARMA(p, q) model

$$y_t = \phi_1 y_{t-1} + \cdots + \phi_p y_{t-p} + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q}$$

Usually, in combination the p and q are smaller than in single AR or MA models

Autocorrelations not particularly useful in practice. For example, for an ARMA(1,1) we have for $k = 1, 2, 3, \dots$

$$\rho_k = \frac{\phi_1^{k-1} (1 + \phi_1 \theta_1) (\phi_1 + \theta_1)}{(1 + 2\phi_1 \theta_1 + \theta_1^2)}$$

What to do in practice?

Check if there is a cut off point in autocorrelations

→ MA model

Check if there is a cut off point in partial autocorrelations

→ AR model

No cut off points?

→ ARMA models

Some variants

ARMAX model

$$y_t = \phi_1 y_{t-1} + \cdots + \phi_p y_{t-p} + \beta_0 x_t + \cdots + \beta_k x_{t-k} + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q}$$

Autoregressive distributed lag model, ARDL(p, k)

$$y_t = \phi_1 y_{t-1} + \cdots + \phi_p y_{t-p} + \beta_0 x_t + \cdots + \beta_k x_{t-k} + \varepsilon_t$$

Note that for forecasting it can be more convenient to consider

$$y_t = \phi_1 y_{t-1} + \cdots + \phi_p y_{t-p} + \beta_1 x_{t-1} + \cdots + \beta_k x_{t-k} + \varepsilon_t$$

ARIMA(p, d, q)

$$y_t = \phi_1 y_{t-1} + \cdots + \phi_p y_{t-p} + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q}$$

where

$$y_t = x_t - x_{t-1} = \Delta_1 x_t$$

which is integrated of order 1, needs to be differenced (d) once

Or perhaps

$$y_t = (x_t - x_{t-1}) - (x_{t-1} - x_{t-2}) = \Delta_1 x_t - \Delta_1 x_{t-1} = \Delta_1^2 x_t$$

where x_t is integrated of order 2

What to do in practice?

First decide on the degree of differencing.

When $y_t = \ln(x_t)$, with $\ln$ is “natural logarithm”, then $\Delta_1 y_t$ is a measure of growth, because

$$100 * (y_t - y_{t-1}) = 100 * (\ln x_t - \ln x_{t-1}) = 100 * \ln \frac{x_t}{x_{t-1}} \approx 100 * \left(\frac{x_t - x_{t-1}}{x_{t-1}} \right)$$

Seasonal ARIMA

Differencing:

$$\Delta_{12}y_t = y_t - y_{t-12}$$

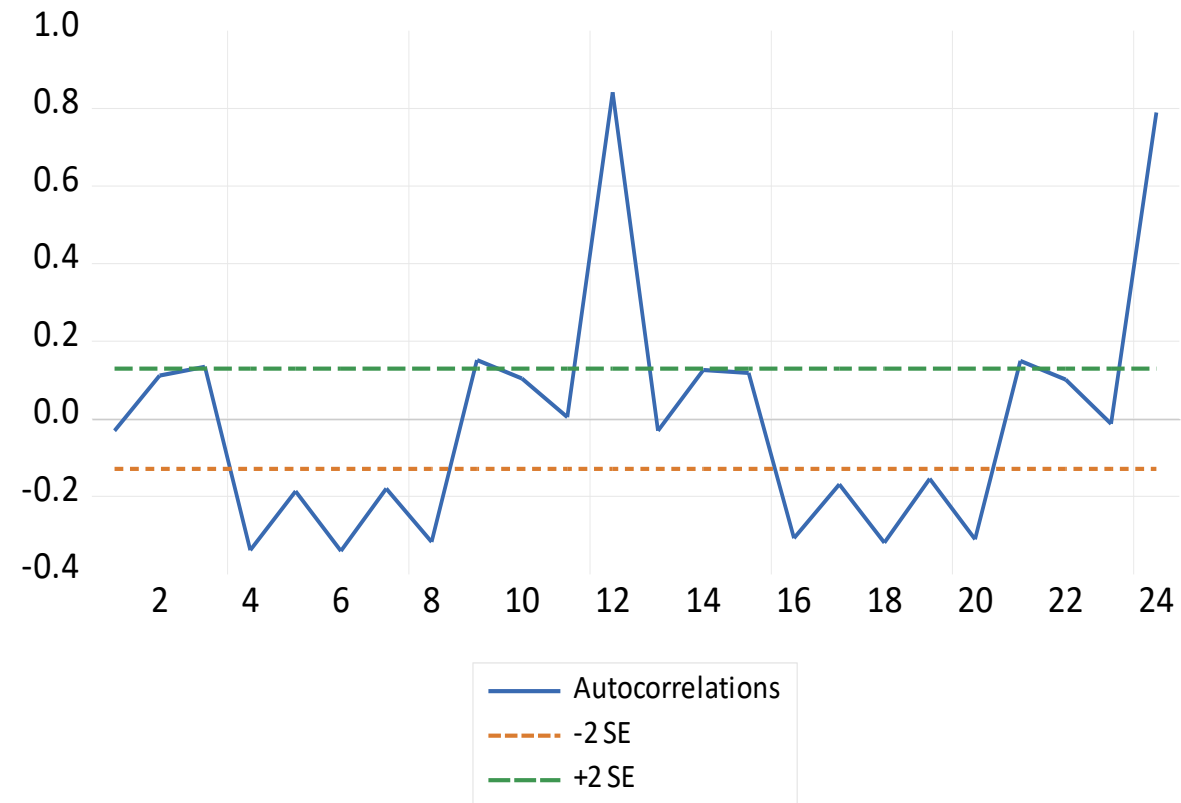
(if months). In general

$$\Delta_S y_t = y_t - y_{t-S}$$

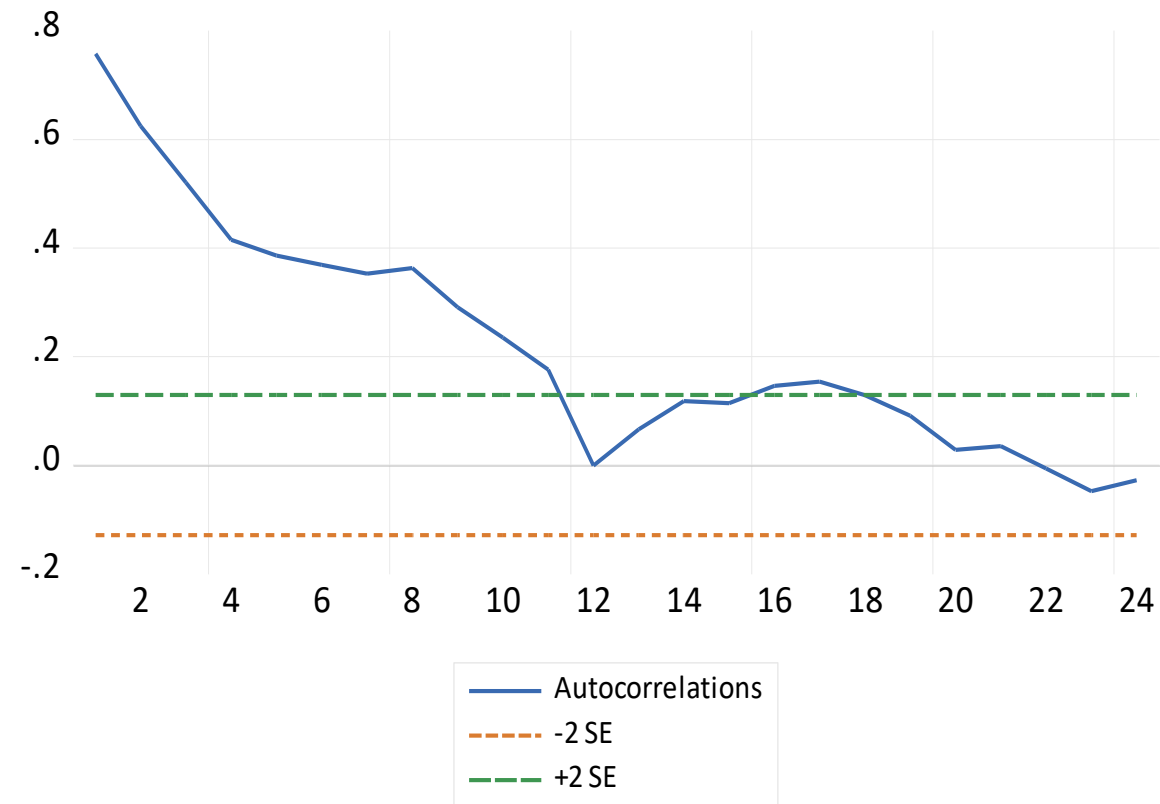
Sometimes:

$$\Delta_1 \Delta_S y_t = y_t - y_{t-1} - y_{t-S} + y_{t-(S+1)}$$

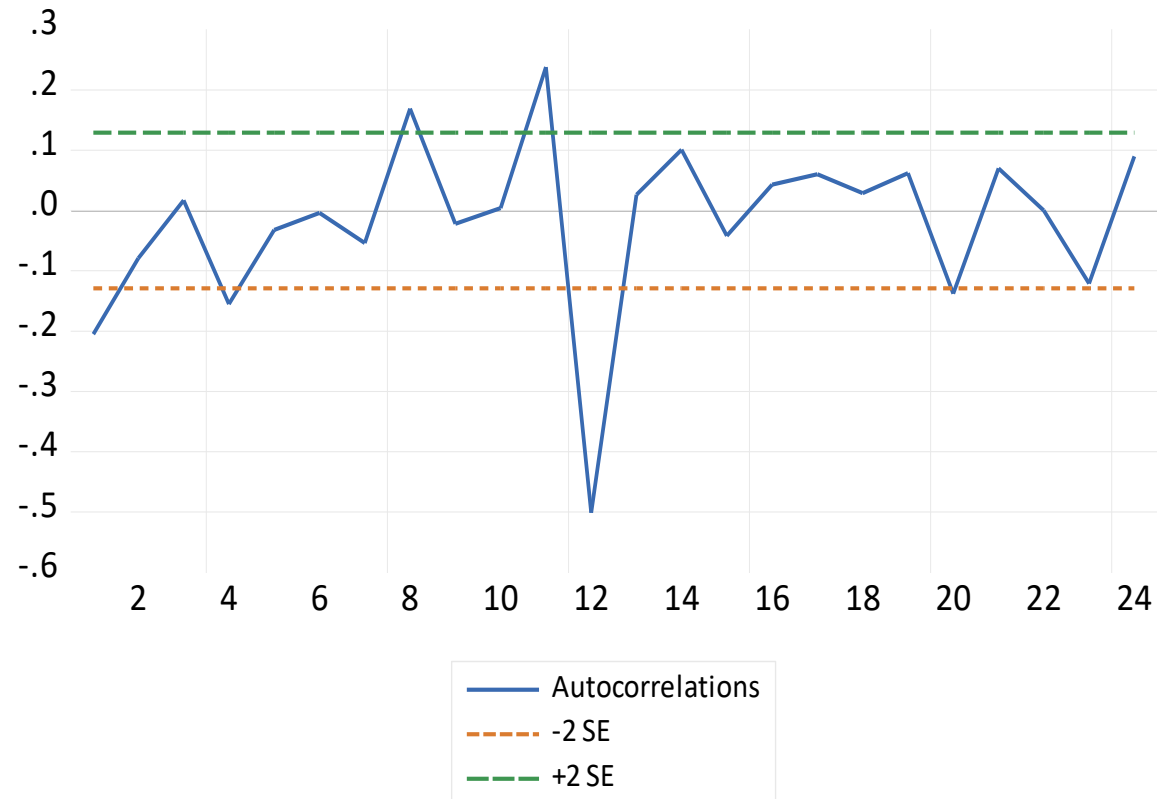
Autocorrelations monthly growth airline revenues



Autocorrelations annual growth rates airline revenues



Autocorrelations of double differenced airline revenues



Estimation results

$$\Delta_1 \Delta_{12} \ln(revenues_t) = \mu + \theta_1 \varepsilon_{t-1} + \theta_{12} \varepsilon_{t-12}$$

	Estimate	SE	t value	p value
C	-0.00038	0.00033	-1.152	0.250
MA(1)	-0.203	0.031	-6.467	0.000
MA(12)	-0.711	0.054	-13.284	0.000

What did we learn?

Autoregression (of order p) and how to identify using the partial autocorrelation function

Moving average model (of order q) and how to identify using the autocorrelation function

ARMA(p, q), ARIMA(p, d, q), and Seasonal ARIMA, and various variants.

No single model for all cases. Each time different.